A Comparative Study of Imbalanced Data Handling Techniques for Detecting Fraudulent Credit Card Transactions

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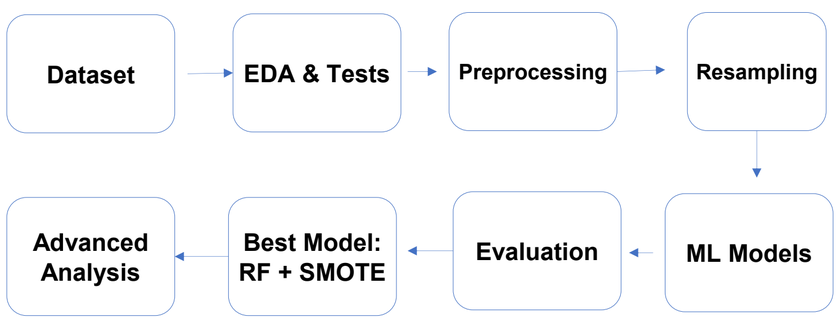
Aims & Objectives

Aim:
To develop and evaluate a machine learning framework for credit card fraud detection by incorporating class imbalance handling techniques in a highly imbalnced dataset.

Objectives:
- Compare different sampling techniques, including SMOTE, Random Oversampling, and Undersampling, across selected machine learning models.
- Evaluate model performance using appropriate metrics for imbalnced fraud dataset.
- Further enhance the performance of the best-performing model using SMOTE-Tomek, ADASYN, cost-sensitive evaluation, decision threshold optimisation and SHAP explainability.

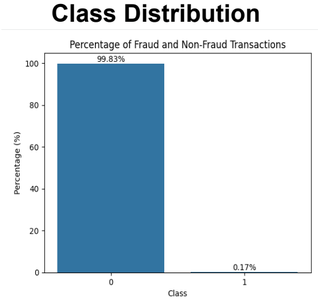
Methods

Dataset:
Credit card fraud detection dataset containing 283,726 transactions, with 473 fraud (0.17%) and 283,253 legitimate (99.83%) transactions.
Data Preparation:
Duplicate removal → stratified 80:20 trian-test split → feature scaling.
Statistical Analysis:
Two-sided exact binomial test and Mannn-Whitney U test.
Modelling:
Logistic Regression, Random Forest, and XGBoost were evaluated using Random Oversampling, Random undersampling, and SMOTE.



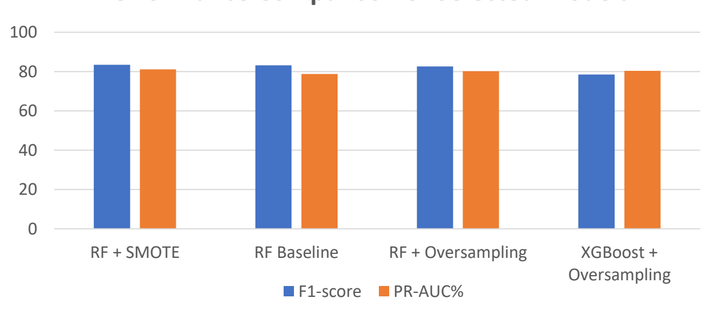
Results

Class Distribution



The dataset showed severe class imbalance, with fraudulent transactions accounting for only 0.17% of all transactions, supporting the need for imbalance-handling techniques.

Performance Comparison of Selected Models



RF + SMOTE achieved the highest F1-score (83.43%) and PR-AUC (0.811) among the selected models and was therefore selected for further analysis.

Enhanced Random Forest Comparison

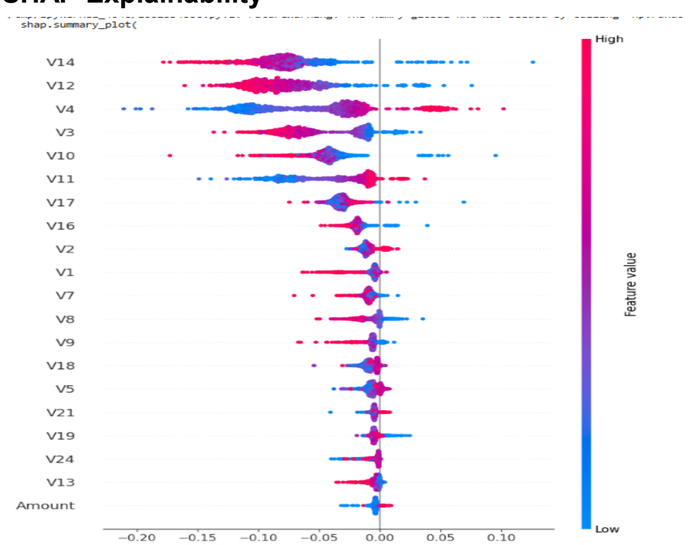
Model	Precision	Recall	F1-score	Cost
RF + SMOTE	91.25%	**76.84%**	83.43%	£11,070
RF + SMOTE-Tomek	**94.74%**	75.79%	**84.21%**	£11,580
RF + ADASYN	89.74%	73.68%	80.92%	£12,580

SMOTE-Tomek achieved the highest F1-score (84.21%), while the original RF + SMOTE produced the highest recall (76.84%) and lowest estimated cost (£11,070)

Decision Threshold Optimisation
Threshold **0.50 → 0.12**
Estimated Cost **£11,070 → £9,040**
≈ 18.3% cost reduction

Key Findings & Explainability

SHAP Explainability



SHAP analysis identified V14, V12, V4, V3 and V10 as the most influential features in the Random Forest model, with V14 and V12 showing the greatest overall impact on predictions.

Key Findings
- Severe class imbalance was confirmed, with fraud representing only **0.17%** of transactions.
- **RF + SMOTE** provided the strongest overall balance in the initial model comparison.
- **SMOTE-Tomek** achieved the highest F1-score (**84.21%**), while RF + SMOTE achieved higher recall and lower estimated cost.
- Threshold optimisation reduced estimated cost from **£11,070 to £9,040 (≈18.3%)**.

Conclusion:
The study demonstrates that class imbalance handling can improve fraud detection beyond what is suggested by accuracy alone. Random Forest with SMOTE provided a strong balance between fraud detection performance and estimated cost, while threshold optimisation further improved its practical effectiveness.

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